

01 — System Overview

1. Problem

Homeowners planning exterior renovation face three barriers before construction:

Gap	Effect
Visualization	They cannot see how <i>their</i> house will look with new cladding, paint, or tiles
Material selection	Choice of finish, location, and durability is guesswork or brochure-based
Cost uncertainty	Quotations vary; area, wastage, and labour are opaque

The product is a **pre-construction planning assistant**: upload a photo, choose finishes, see a redesign of the actual house, and get an approximate quantity and cost breakdown suitable for discussion with contractors.

2. Goals (from problem statement)

1. Accept exterior images of a residential building
2. Generate redesigned visual options using selected materials
3. Estimate material quantities required
4. Calculate a transparent renovation cost

3. Target users

Role	Use
Primary	Homeowners, new house owners before finishing
Secondary	Contractors, architects, builders, consultants, material suppliers

4. Scope of this prototype

In scope

- Single exterior photo upload with quality guidance
- AI segmentation of major facade categories

- Manual mask refinement (brush / eraser)
- Material catalog (paint + textures: cladding, tiles, patterns)
- Classical CV rendering preserving structure and lighting
- Before / after comparison
- Surface quantity from mask coverage $\times$ user facade size
- INR cost estimate with editable rates
- Downloadable PDF report

Out of scope (by design)

- Interior design
- Structural / engineering calculations
- Professional CAD drawings or survey equipment
- Legally binding quotations (estimates are **advisory**)
- Multi-photo fusion, NeRF / full 3D reconstruction
- User authentication / multi-tenant accounts (single-user local demo)

5. Assumptions (aligned with problem statement)

- Residential, low-rise (independent house, bungalow, small apartment)
- Exterior renovation only
- User can approximate facade width and height in metres
- One photo shows the primary elevation of interest
- Cost estimates are for planning discussion, not contracts

6. Constraints met

Constraint	How we satisfy it
No architectural drawings required	Photo-only input
No specialised measuring equipment	User facade W $\times$ H + mask fractions
Rely primarily on visual media	Upload $\rightarrow$ segment $\rightarrow$ render pipeline
Usable by non-technical users	Guided studio UI, category buttons, cost page

Reasonable time on standard connections	CPU classical render (~seconds); segmentation slower on first load
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7. Solution shape (one sentence)

AI perceives regions; deterministic math costs them — segmentation and rendering produce a RegionMap; quantity and cost are pure formulas over that map plus editable rates, never opaque model guesses for money.